All credit and glory to God. I owe my knowledge to my Lord Jesus Christ.

MindFrame: How to Test It, and What Could Come Next

A validation plan for the MindFrame model, and a concept for a lifelong growth app that carries its legacy

Prepared for Ryan Espinosa (GitHub **ryryespi**) · October 19, 2026

> **Model, not validated science.** MindFrame is a model built from design assumptions and informed by research. It is not therapy, not a diagnosis, and it has not been tested in people. Nothing in this document claims that it works; this document is about how to find out honestly. In a crisis in the US, call or text **988**, or call **911**.

In one page

What MindFrame is today. MindFrame (`github.com/ryryespi/mindframe`, live at `ryryespi.github.io/mindframe`) is a private, offline web app (a PWA). You write one thought, or pick feelings, and tap Update. The app reads the words into nine "movers" and takes one gentle step toward a home state. Everything stays in the browser. The README and the papers in `paper/` already say plainly: "Not validated in people. No user study, clinical trial, or measurement study has been done." The Grace–Peace paper names the next step: "a small, ethically reviewed user study." This document is a concrete version of that step.

Part 1 lays out three phases, each with a gate you must pass before going on:

1. **Phase 1, a study of one (you).** 10 weeks, ABAB design, daily check-ins inside the app, weekly standard questionnaires. Every new model piece gets a prediction that could fail.
2. **Phase 2, a small pilot (about 20–40 people)** with ethics review, consent, a safety plan and on-device data. It checks whether people can use it safely, and whether the in-app questions measure what they claim.
3. **Phase 3, only if warranted:** a randomized comparison against journaling only.

Part 2 imagines the app that comes after, assuming the evidence is good. I call it **Trellis**: a lifelong growth garden. Your history becomes a map of seasons, the model is calibrated to you, an optional faith layer adds the Examen and spiritual disciplines, and sharing with a counselor or pastor is always your choice. It ends with a build roadmap you can use to keep learning to code, one milestone at a time.

The model being tested (from the repo)

```
e ∈ [0,1]⁹ : fear, sadness, anger, shame, calm, connection, hope, rumination,
avoidance
home h : calm = connection = hope = 0.8, all others 0
step   : e' = clip(e + a(e)(h − e) + b·s(e)),  a(e) = min(a_max, a0 + k‖e − h‖)
peace  : P = P_d(e)·(1 − γC)·(1 − δG),  P_d = 1/(1 + (d/d0)²),  d = ‖e − h‖
shield : S = 1 − (1 − A)(1 − F)(1 − E)(1 − R)   (Kai: self-awareness, forbearance,
empathy, respect guard)
baseline (already in code): B_t = 0.8·B_(t−1) + 0.2·e_t
```

The pieces being added now are a layered topology with a moving **anchor chain**, **per-emotion time constants**, **resistance ρ**, an **expression governor** (a control system that watches for unawareness, annoyance, and mania-like activation), story-following images, and personalization by hobbies and field. These are not in the public repo yet (I checked the code on October 19, 2026), so this plan writes them in a general form. **Before Phase 1 starts, freeze their exact equations and default values in the pre-registration.** That way the predictions are fixed before the data come in.

Part 1 — Validation plan

What "validation" means here

There are three different questions, and it helps to keep them apart:

1. **Does the model describe you?** Do its numbers track what you actually feel (for example, does P go up when you feel more at peace)? Do its dynamics match (does sadness really fade more slowly than fear)?
2. **Can the model predict?** Does the anchor chain predict tomorrow better than a simple guess? Does the governor warn before irritable days?
3. **Does using the app help?** Are your days calmer, more self-compassionate and less ruminative with the app than without it?

The math proof in the paper (the step is a contraction toward h) is about the model's arithmetic only. As the paper itself says, "It says nothing about real emotions." Only data can answer the three questions above.

Phase 1 — A single-case (N-of-1) study on Ryan

Why start with one person. Single-case designs are a recognized method when each phase is repeated and the data are dense (Kazdin, 2011; Kratochwill et al., 2013). N-of-1 trials have a practical guide from the US Agency for Healthcare Research and Quality (Kravitz & Duan, 2014). There is also a deeper reason that fits MindFrame: findings averaged over groups often do not describe any one person well (Fisher, Medaglia & Jeronimus, 2018). A personal model should first be tested on a person.

Design: ABAB, 10 weeks (range 8–12)

Phase	Length	What you do
A1 baseline	2 weeks	Daily check-in and a short plain journal entry. The app's model runs in **shadow mode**: it computes e, P, S, anchors and governor alerts and logs them, but shows no card, step or practice.
B1 MindFrame	3 weeks	Same check-in, plus full MindFrame use (Update, card, one step, practices).
A2 withdrawal	2 weeks	Back to check-in + plain journal, model in shadow mode.
B2 MindFrame	3 weeks	Full use again.

- **Why ABAB.** It gives three chances to see the effect turn on and off. Published standards ask for at least three demonstrations of an effect, at least four phases, and at least five data points per phase (Kratochwill et al., 2013). Daily data give 14–21 points per phase, which is plenty.

- **Randomize the switch days.** Before you start, let a trusted friend (a "data steward") randomly pick each switch day from a window (for example, day 12–16 for A1→B1). That makes a randomization test possible (Edgington & Onghena, 2007), and it keeps you from switching on a "good" day.
- **Honest catch.** Learning does not always switch off. If MindFrame teaches you a skill in B1, A2 may not fall back, and that would hide a real effect. If this worries you, use a **multiple-baseline** design instead: introduce MindFrame's parts one at a time (for example the step card, then the Kai shield, then the practices) on staggered, randomized start days.
- **Shadow mode is key.** Because the model runs in every phase, you can test the model's predictions (questions 1 and 2) on all 70 days, not just the B weeks.

Measures

Measure	When	What it is and why
SUDS, 0–100	Daily (evening), plus before/after any Update	Subjective Units of Distress, from behavior therapy (Wolpe, 1969). One quick number for "how distressed am I right now." The before/after pairs show short-term change.
Short affect items	Daily	The 10-item international short form of the PANAS (Thompson, 2007), from the original PANAS (Watson, Clark & Tellegen, 1988). Gives positive and negative affect scores that do not come from MindFrame's own words.
Nine mover items, 0–10	Daily	One plain item per mover ("How afraid have I felt today?"), plus "How at peace do I feel?", "How irritable?", and "How energized or revved up?". These are the in-app items whose quality Phase 2 checks.
Sleep	Daily (morning)	Hours slept, time to bed, time up. Items can follow the Consensus Sleep Diary (Carney et al., 2012). Sleep matters for mood and for the governor.
SCS-SF	Weekly	Self-Compassion Scale–Short Form, 12 items (Raes, Pommier, Neff & Van Gucht, 2011). The total score correlated $r \geq .97$ with the long form. Use the total only: its two-item subscales are not reliable.
AAQ-II	Weekly	Acceptance and Action Questionnaire–II, 7 items (Bond et al., 2011). Measures experiential avoidance and psychological inflexibility. It lines up with the *avoidance/fusion* mover, which already cites it.
RRS Brooding	Weekly	The 5-item Brooding subscale of the Ruminative Responses Scale (Treynor, Gonzalez & Nolen-Hoeksema, 2003). It is the part of rumination most tied to distress, and it checks the *rumination* mover.
PHQ-9, **GAD-7**	Weekly	Standard depression (Kroenke, Spitzer & Williams, 2001) and anxiety (Spitzer et al., 2006) questionnaires. A score of 10 or more is the usual flag (for PHQ-9, 88% sensitivity and 88% specificity in the original study). Here they serve as **safety and harm** checks, not as a diagnosis.
Peace of Mind	Weekly	The Peace of Mind scale (Lee, Lin, Huang & Fredrickson, 2013), the construct MindFrame's P already cites. It is an outside check on P.
Activation & irritability (optional)	Weekly	Altman Self-Rating Mania Scale (Altman et al., 1997) and Brief Irritability Test (Holtzman et al., 2015). Outside checks for the governor. Treat them as signals to look at, never as a diagnosis.

Check each scale's permission terms before you build it into an app; some are free for research or clinical use but have copyright notes. Keep the check-in under about 3 minutes, because long daily surveys wear people out (Shiffman, Stone & Hufford, 2008).

Hypotheses: one testable prediction for each model piece

Each prediction is written so it can fail. "Falsified" here means the piece did not earn its place *for you*. In that case you recalibrate it, simplify it, or remove it, and you report it either way.

Model piece	Prediction	Counts as falsified if…
H1 Time constants τ_i	After a spike, each mover fades toward home at its own pace: $e_i(t) - h_i \approx (e_i(0) - h_i)\cdot\exp(-t/\tau_i)$. The model's pre-set τ values predict the observed half-lives. Published work gives an outside expectation too: in recalled episodes, sadness lasted longest of 27 emotions, and shame and fear were among the shortest (Verduyn & Lavrijsen, 2015).	Fewer than 2 in 3 of the movers that have enough spikes (at least 4 each) have the predicted τ inside the 95% interval of the observed decay, *or* the predicted and observed rank order correlate below .3.
H2 Anchor chain	Today's anchor predicts tomorrow's nine mover ratings better than (a) "tomorrow = today" and (b) your average so far.	On held-out days (the last 3 weeks), the anchor's mean absolute error is not at least 10% lower than *both* simple guesses.
H3 Resistance ρ	Higher ρ on a day means slower recovery: the effective step is about $a\cdot(1 - \rho)$, so recovery time (SUDS back under your median) grows with ρ.	The within-person slope of recovery time on ρ is not positive, or its 95% interval includes zero.
H4 Expression governor	An alert on day t comes before higher irritability (or activation) ratings on day t+1, compared with no-alert days, after accounting for today's rating and sleep.	Precision below .5, more than one false alarm a week on average, or no lagged effect (interval includes zero).
H5 Inner peace P	P from your words tracks your own daily "How at peace?" rating and the weekly Peace of Mind score.	Within-person correlation with the daily rating is below .30, or its interval includes zero.
H6 Kai shield S	On days with outside criticism ($X > 0$), higher S goes with a smaller rise in anger and shame ratings.	No negative slope (expect few such days; report as exploratory if fewer than 8).
H7 Using the app	SUDS and negative affect are lower, and peace higher, in B than in A phases, at all three transitions.	The randomization test is not significant ($p \geq .05$), or Tau-U is small (below .2), or the effect shows at fewer than 3 transitions.
Harm check	No worsening.	A reliable increase in PHQ-9 or GAD-7 (reliable change index, Jacobson & Truax, 1991) during B phases. If so, stop and talk with someone you trust.

Story-following images and hobby/field personalization are about engagement and meaning, not about emotion dynamics. In Phase 1, track only whether you use them and a 0–10 "this felt like my story" rating. Don't make claims about them yet.

Analysis

- **Look first.** Plot every daily series with the phase lines. Visual analysis is standard in single-case work (Kazdin, 2011). Report it using the SCRIBE checklist (Tate et al., 2016).
- **Single-case effect size: Tau-U.** It combines non-overlap between phases with control for baseline trend (Parker, Vannest, Davis & Sauber, 2011).
- **Randomization test.** Because the switch days were picked at random, compare the observed A-vs-B difference with every possible switch schedule (Edgington & Onghena, 2007). This gives an honest p-value for one person.

- **Time-series / multilevel models.** Model the daily ratings with an AR(1) term (each day depends on the day before), phase, and sleep. Treat the measurement occasions as nested in days (Bolger & Laurenceau, 2013). The AR coefficient is also a direct estimate of "emotional inertia", which in published work was higher in people with more depressive symptoms (Kuppens, Allen & Sheeber, 2010). That gives you a natural check on the time constants.
- **Calibration plots** for anything the model predicts: plot predicted against observed in bins, and report the calibration slope and intercept. Good average accuracy can hide bad calibration (Van Calster et al., 2019).
- **Pre-registration on OSF.** Before day 1, post the hypotheses, the frozen model equations and defaults, the measures, the switch windows, the analysis script, and the falsification rules (Nosek, Ebersole, DeHaven & Mellor, 2018). Changes after that are allowed, but you label them "exploratory".
- **Hold-out data.** Fit nothing on the last 3 weeks; use them only to test H2 and H4.

> **Honest limits of Phase 1.** You built the app, you hope it works, and you can't be blinded. Expectation and the wish for a good result can move self-ratings. You also can't generalize from one person: a positive result means "promising for Ryan", not "it works." Some things reduce the bias: pre-registration, randomized switch days held by someone else, automated logging, standard questionnaires next to the app's own items, and writing the analysis before seeing the data. They reduce it; they don't remove it.

> **Your own safety in Phase 1.** Before you start, write a short safety plan (warning signs, coping steps, people to contact, professional contacts, 988/911), following the Safety Planning Intervention (Stanley & Brown, 2012). Tell one trusted person and, ideally, a counselor or pastor that you are doing this. Agree ahead of time: if PHQ-9 item 9 is above 0, or any weekly score reaches 15 or more, you pause the study and reach out. Being well matters more than the data.

Gate to Phase 2: no harm signal; at least H5 and H7 supported; and the in-app items behave sensibly (they vary, they aren't always at the floor or ceiling, and they relate to the standard scales). If the model pieces (H1–H4) fail, simplify the model before you recruit anyone. A smaller honest model is better than a bigger untested one.

Phase 2 — Small pilot (n ≈ 20–40)

Purpose: feasibility, safety and measurement quality. It is *not* meant to prove the app works. Pilot studies are too small to give reliable effect sizes, and their effect sizes should not be used to plan a larger trial (Leon, Davis & Kraemer, 2011). A common rule of thumb is at least 12 people per group for a pilot (Julious, 2005). With 20–40 people you can estimate recruitment, retention, adherence and item reliability with useful precision.

- **Ethics review (IRB).** Research with people needs review by an IRB or ethics board. As an independent learner, the realistic path is to partner with a faculty member at a university or seminary, or a clinic, that has an IRB. Commercial IRBs also exist. Do not recruit anyone before approval.
- **Informed consent** in plain words: what the app is and isn't ("model, not validated science; not therapy"), what data are collected, that participation is voluntary and you can stop anytime, the risks (feeling worse while reflecting, privacy), and who to contact.
- **Who:** adults 18+. Exclude people in current crisis, with recent suicidal behavior, or with a known bipolar or psychotic disorder unless a clinician is co-supervising. Screen with PHQ-9 and GAD-7,

and refer anyone scoring high to care.

- **Safety plan:** the 988 line always visible (already in MindFrame); the crisis-wording box (already in MindFrame); a named study clinician; a rule for weekly PHQ-9 item 9 > 0 (the participant sees 988/911 right away, and the study contact follows up within one business day); an adverse-event log; stopping rules.
- **Data privacy:** data stay on the device, as MindFrame does now. For the study, add an **opt-in anonymized export**: the participant taps Export, sees exactly what will be sent (numbers, item scores, timestamps rounded to the day, never their words), and sends a file under a random study ID. The key that links IDs to names is kept apart, by the study team only.
- **Design:** 4–6 weeks of daily check-ins plus app use. You can repeat a short A/B block within each person so each participant is also a small N-of-1 study, and the results can be combined across people with multilevel models.

Psychometric checks of the in-app items

Check	How
Reliability	Within-person and between-person reliability of the mover items from multilevel models (Bolger & Laurenceau, 2013). Test–retest for the weekly scales.
Convergent validity	The fear and sadness items should relate to PANAS negative affect and GAD-7/PHQ-9; rumination to RRS Brooding; avoidance to AAQ-II; P to Peace of Mind; the shame and self-talk signals (inversely) to SCS-SF. Write down the expected direction and rough size before you look.
Discriminant validity	The movers should not all be one "bad day" factor. Check the correlations between movers and the factor structure.
Text-reader accuracy	Compare the keyword reader's nine scores with the same person's own ratings, and with a sample of entries labeled by two trained raters (only with explicit consent to share text). Report misses and false alarms, including for crisis wording. The paper already lists this as future work.
Usability and burden	Completion rate, minutes per day, drop-out, and short interviews. For context: in one large analysis of popular mental-health apps, median 30-day retention was 3.3% (Baumel et al., 2019).

Gate to Phase 3: at least 70% of people finish; no serious harm; in-app items with acceptable reliability and the expected relations to the standard scales; and the individual-level results from Phase 1 mostly hold up across people.

Phase 3 — Randomized comparison (only if warranted)

- **Arms:** MindFrame vs. **journaling-only** (the same app shell and daily check-in, a plain writing box, no model output). Journaling is a fair, active comparison: expressive writing has small but real benefits on average (Frattaroli, 2006). Beating it means the model adds something beyond writing things down.
- **Primary outcome** chosen and registered in advance (for example, change in PHQ-9 or in Peace of Mind at 8 weeks), plus follow-up at 3 months.
- **Sample size** from the smallest difference that would matter to people, *not* from the pilot's effect size (Leon et al., 2011). Expect a few hundred participants; that needs a research partner and probably funding.
- **Keep expectations humble.** A meta-review of 14 meta-analyses of randomized trials found the evidence for phone-based mental-health interventions to be modest and of limited quality

(Goldberg et al., 2022). A small, honest effect would already be a good outcome.

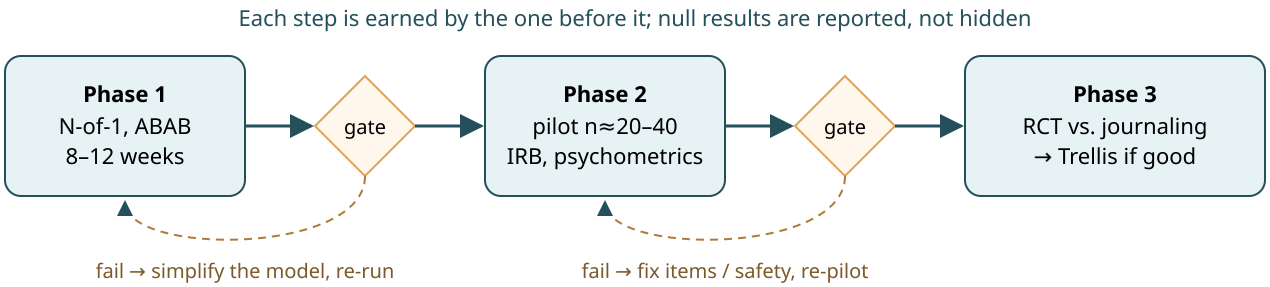


Figure 1. The three phases and their gates.

Part 2 — What comes next: *Trellis*

This part assumes the validation goes well. If it doesn't, most of Trellis still makes sense with fewer model claims (see "If the evidence is weak" below).

Trellis — a lifelong growth garden.

Trellis is a private, on-device reflection companion that turns your days into a map of seasons, so you can see where you have been, what helped, and what is slowly growing. It carries MindFrame's legacy: one gentle step at a time, words before numbers, and a model honestly calibrated to *you* rather than to an average person. With an optional faith layer (the Examen, spiritual disciplines, Scripture), and sharing that is always your choice, it helps you grow in peace, values and good habits over years, not just get through a hard day.

Why the name. A trellis doesn't make a plant grow. It gives the growth something steady to lean on and a shape to follow. That is the honest role of an app. It also fits the picture already at the heart of MindFrame's cornerstone scene: *"He will be like a tree planted by the streams of water, that produces its fruit in its season"* (Psalm 1:3, WEB). The word works with faith on or off. (Working name only. A pregnancy-care app called Trellis Health exists, so do a name and trademark check before you publish.)

What carries over from MindFrame

MindFrame today	Becomes in Trellis
Nine movers, the step to home, P	The same core, with parameters (h, τ_i, ρ, weights) **calibrated per person** from their own check-ins, and shown with uncertainty ("we're still learning your pace").
Baseline B_t and the anchor chain	The **life map of seasons**: long stretches of your anchor chain are grouped into seasons you can name.
Kai shield, practices, forgiveness	A **practice garden**: the practices that actually helped *you* (by your own before/after SUDS) move to the front.
Faith toggle (words change, math doesn't)	An optional **faith layer** with the daily Examen and spiritual disciplines; still never changes the math.
Local-only, no account, mf code	A **privacy-first** architecture: encrypted on the device, with export, sharing and groups all opt-in.

Story images, hobbies/field	Your story told in the pictures and language of your own world (a coder's debugging season, a musician's practice season).

The parts of Trellis

1. The life map of seasons

The anchor chain is a slow-moving summary of your state. Trellis looks for long, stable stretches and the turning points between them (a change-point method on the anchor chain), and offers them as **seasons**. For example: a *winter* (low hope, high sadness, rest needed), a *spring* (hope rising, new practices), a *summer* (steady and full), a *harvest* (growth you can name). You name and edit them; the app only suggests. Each season keeps what helped, the people who were there, and one sentence you want to remember. Over the years, this becomes a map you can look back on with gratitude, instead of a feed of scores.

2. A model calibrated to you

Start every user with MindFrame's defaults, then slowly fit their own home, time constants and resistance from their check-ins, using the same methods that passed in Phase 1. Show it in plain words ("your sadness tends to ease over about three days; fear passes within the day") and always with honesty about uncertainty. The calibration is checked continuously (calibration plots, inside the app's "Show the math"). When the model is poorly calibrated for someone, it says so and leans on their own ratings instead.

3. Optional faith layer

- **The daily Examen** from the *Spiritual Exercises* of St. Ignatius of Loyola. In Ganss's translation (1992, §43), the general examen has five points: give thanks to God for benefits received; ask grace to know your sins and be rid of them; review the day, hour by hour, in thoughts, words and deeds; ask pardon for faults; and resolve, with God's grace, to amend. It closes with the Our Father. Trellis can offer it as a 10-minute evening practice, with the day's seasons and movers as gentle prompts ("Where did you notice peace today? Where did anger rise?").
- **Spiritual disciplines.** Richard Foster's *Celebration of Discipline* (1978) groups twelve disciplines as inward (meditation, prayer, fasting, study), outward (simplicity, solitude, submission, service) and corporate (confession, worship, guidance, celebration). Each can be a small, trackable rhythm in the habit garden.
- Scripture keeps MindFrame's rule: exact World English Bible text, references shown, faith wording never changes the numbers.

4. Values and purpose

Following the values work in Acceptance and Commitment Therapy (Hayes et al., 2006), you name a few values (for example, faithfulness, craft, family, service), and each season and habit is tied to one. The question shifts from "How do I feel less bad?" to "What am I growing toward?". That is MindFrame's home h, chosen by the person.

5. Habit formation

- **Plan with if–then cues.** "After I pour my morning coffee, I will pray the Examen for five minutes." Implementation intentions like this help people follow through (Gollwitzer, 1999).
- **Honest timelines.** In a real-world study of 96 volunteers, a daily behavior took a median of 66 days to reach its plateau of automaticity, with a range of 18 to 254 days. Missing one day did not materially affect habit formation (Lally, van Jaarsveld, Potts & Wardle, 2010; see also Gardner, Lally

& Wardle, 2012). Trellis says this out loud: no “21-day” promises, no streak shame, and a missed day is just a missed day.

- Show automaticity growing as a plant on the trellis, not as a streak counter.

6. Counselor / pastor share mode (user-controlled)

You choose what to share (for example, the last 4 weeks of seasons and mover trends, never journal text unless you pick specific entries), and with whom. You preview the summary, then hand it over as a file or printout, or show it on your screen in the session. In the first version there is no server and no standing access, so there is nothing to revoke later. A counselor sees a summary *you* made.

7. Small-group anonymous mode

For a church small group or recovery circle. The group shares a weekly prompt and a practice. Members can contribute only coarse, anonymous signals (“this week was a winter / spring”, “I tried the practice: yes/no”), shown only when at least 5 people answered. No free text, no names, no individual trends. Be honest in the app: in a group of 6 friends, anonymity is fragile, so share only what you would be fine being guessed.

8. Privacy-first architecture

Follow the “local-first” ideals: your data live on your device, the app works offline, and you own your data (Kleppmann et al., 2019). Concretely: data encrypted at rest with a key from your passphrase; no account; no analytics; every network call is shown before it happens (as MindFrame already does with Wikidata); research export is opt-in and anonymized. Groups need a small relay server, and it should only ever see encrypted, aggregate data.

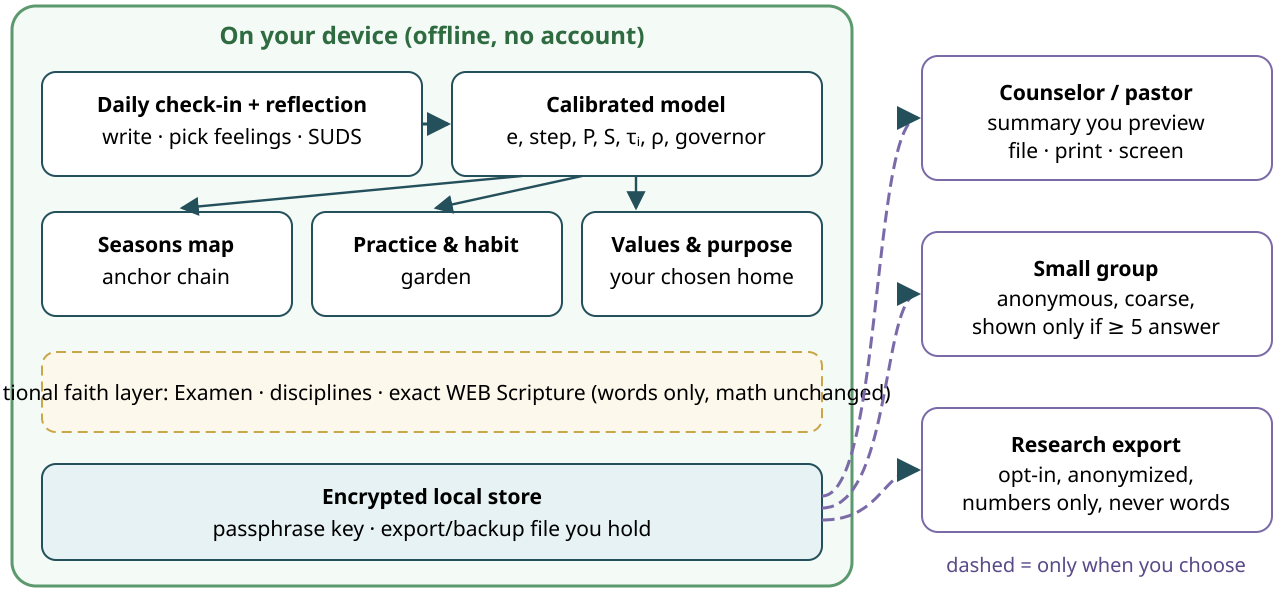


Figure 2. Trellis architecture: everything lives on the device; every arrow leaving it is opt-in.

Build roadmap: learn by building

Each milestone is a real, shippable step, and each teaches one new layer of software development. Keep MindFrame's habits the whole way: tests for every model change, a header comment on every file, and the "model, not validated science" label until evidence says otherwise.

#	Milestone	What you build	What you learn	Done when
M0	Finish MindFrame's new layer	Anchor chain, τ_i, ρ, governor in `js/core/`, each with node tests and README math	Modeling in code, unit tests, writing specs	Equations frozen and pre-registered on OSF
M1	Study mode in MindFrame	Daily check-in screen, weekly scales, shadow mode, CSV export	Forms, IndexedDB, schema versions, data export	Phase 1 can start
M2	Analysis notebook	Python (pandas, statsmodels) or R scripts: plots, Tau-U, randomization test, AR(1), calibration	Data analysis, reproducible scripts, Git for research	Script runs on fake data before real data exist
M3	Phase 1 report	Results write-up with SCRIBE checklist, nulls included	Scientific writing, honesty under uncertainty	Report posted with the repo
M4	Trellis skeleton	New repo: TypeScript + Vite, a light UI library (Svelte or Preact), PWA, MindFrame core ported with all tests	TypeScript, build tools, components, porting tests	Same numbers as MindFrame on every test
M5	Seasons map	Anchor-chain change points, SVG season timeline, naming and editing seasons	Algorithms on time series, SVG/visualization, accessibility	You can see and name your own seasons
M6	Calibration engine	Per-user fitting of h, τ_i, ρ in a Web Worker, with uncertainty and in-app calibration plots	Optimization, background workers, numerical care	Calibration on your Phase 1 data matches the notebook
M7	Faith, values, habits	Examen flow, disciplines, values picker, if–then habit plans, automaticity plant	UX writing, state machines for guided flows	A full evening Examen works offline
M8	Privacy hardening	WebCrypto encryption at rest (passphrase-derived key), encrypted backup file, threat model doc	Applied cryptography basics, threat modeling	An outside reviewer reads the threat model
M9	Share mode	Preview-then-export counselor summary (PDF/print), with per-item choice	Generating documents, consent-centered design	A counselor or pastor tries it and gives feedback
M10	Pilot build	Opt-in anonymized export, study IDs, safety flows checked by a clinician	Working with an IRB, release discipline	Approved for Phase 2
M11	Small groups	Tiny relay server (for example Node or Deno) holding only encrypted aggregates; minimum group size 5	Backend basics, APIs, end-to-end encryption	Security review passed
M12	App stores (optional)	Wrap the PWA (for example with Capacitor), accessibility audit with screen-reader users	Mobile packaging, store policies for health apps	Published after Phase 3, or with honest "not validated" labels

If the evidence is weak

That would not be failure; it would be finding something out. If the model pieces don't predict well, Trellis can drop the claims and keep the parts that don't depend on them: private journaling, the season map built from your own ratings, the Examen, values, habits with honest timelines, and user-controlled sharing. MindFrame's best legacy may be its manner more than its math: transparent, gentle, private, and honest about what it knows.

> **Limits of this document.** It is a plan, not a result. The new MindFrame pieces are described in general form because their code was not public on October 19, 2026. The thresholds in the hypotheses (.30, 10%, .5 precision) are reasonable starting points that I chose, not field standards; adjust them before pre-registering. This plan is not a substitute for advice from an IRB, a clinician or a statistician, and Phases 2–3 need those partners.

References

Every source below was checked on October 19, 2026: journal articles against Crossref by DOI (authors, year, title, journal, volume, pages), books against Open Library, and the AHRQ guide at its source. Specific numbers quoted in the text (Lally 66 / 18–254 days; Baumel 3.3%; SCS-SF r ≥ .97; PHQ-9 88%/88%; Brooding 5 items; Verduyn & Lavrijsen findings; Kratochwill standards; Ganss §43) were checked against the papers' abstracts or full text.

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Project sources: MindFrame README and papers (`paper/MindFrame-One-Equation.pdf`, `paper/MindFrame-Grace-Peace-Paper.pdf`), github.com/ryryespi/mindframe, commit be33040 (October 19, 2026).